MANIKANDAN SOMU

Geodesy & Geospatial Data Specialist
Remote Sensing — Earth Observation

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SUMMARY

Geodetic Engineer with professional experience in geospatial analysis, scientific programming, and Earth observation. Interested in scientific computing and Earth system applications, particularly in remote sensing, hydrology, and environmental monitoring, with a strong motivation to contribute to innovative research and engineering projects.

SKILLS

Scientific Programming: Python, MATLAB, HTML5, Javascript, SQL

GIS & Spatial Technologies: QGIS, ArcGIS, PyQGIS, GDAL, PostgreSQL/PostGIS, OpenStreetMap

Earth Observation: GRACE(-FO), Sentinel-1, Sentinel-2, Landsat, ERA5, GLDAS, Satellite Altimetry

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EDUCATION

University of Bonn — Bonn, Germany
M.Sc. in Geodetic Engineering — **Oct 2022 - Jun 2026**

College of Engineering Guindy, Anna University — Chennai, India
Bachelor's in Geoinformatics Engineering — **Aug 2015 - Jun 2019**

WORK EXPERIENCE

Working Student – Full Stack Geospatial Developer — **Jun 2024 - Jul 2025**
Navilan Solution GmbH — Bonn, Germany

- Developed full-stack geospatial applications using Python, web technologies, and spatial databases.
- Built and maintained backend services, APIs, and geospatial data processing workflows.
- Developed interactive WebGIS interfaces and geospatial visualization tools.
- Integrated spatial databases and Earth observation datasets into operational applications.

Student Researcher - Geospatial Deep Learning — **Jan 2023 - Dec 2023**
PhenoRob Lab, University of Bonn — Bonn, Germany

- Processed multi-temporal satellite and UAV datasets for environmental and agricultural analysis.
- Applied deep learning methods for spatial feature extraction and remote sensing applications.
- Performed preprocessing, validation, visualization, and interpretation of Earth observation datasets.

Senior GIS Analyst — **Jul 2019 - Oct 2022**
Trimble Inc. — Chennai, India

- Processed and validated large-scale geospatial datasets for mapping and infrastructure-related projects.
- Developed and supported QGIS plugin workflows for spatial data processing and quality control.
- Worked with raster and vector datasets including POI, AOI, and mapping layers.
- Performed data validation, quality checking, and structured GIS data management.
- Supported workflow automation using Python-based geospatial processing tools.

- Worked with coordinate systems and spatial reference frameworks.

PROJECTS

Investigating Mass variability on the Earth Surface Using Satellite Gravimetry **Nov 2025- Jun 2026**
Master's Thesis

- Processed and analysed GRACE(-FO) satellite gravimetry products for regional and global mass change studies.
- Developed MATLAB and Python workflows for spatio-temporal analysis of large geodetic datasets.
- Integrated GRACE observations with satellite altimetry, ERA5 precipitation and runoff, GLDAS terrestrial water storage, and GRDC river discharge datasets.

High-Precision Space Geodesy Simulator **Apr 2024 - Apr 2025**
Research Project

- Developed numerical simulation workflows for satellite trajectory modeling and geodetic analysis.
- Implemented efficient computation methods for high-precision space geodesy applications.

Mesoscale Eddy Detection in the Bay of Bengal **Aug 2018 - Apr 2019**
Bachelor's Thesis

- Investigated the occurrence and movement of mesoscale eddies in the Bay of Bengal using satellite altimetry observations.
- Processed sea surface height anomaly (SSHA) data to identify cyclonic and anticyclonic eddy structures.
- Analysed the spatial and temporal evolution of eddies and their propagation patterns within the study region.
- Evaluated the influence of mesoscale eddies on regional ocean circulation and sea surface dynamics.

Time Evolution of Winter Cooling and Associated Biogeochemical Processes in the Northern Arabian Sea **Jan 2019 - May 2019**
Research Project, National Institute of Oceanography (NIO), India

- Investigated winter cooling processes and their influence on physical and biogeochemical conditions in the Northern Arabian Sea.
- Analysed oceanographic and environmental datasets to study seasonal variability in sea surface temperature and related marine processes.

LANGUAGES

German (A2), English (Fluent), Tamil (Native)